

Impact of Enhanced Seasonal Malaria Chemoprevention on Haemoglobin Trajectories: A Longitudinal RCT Analysis

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Abstract. Malaria-associated anaemia is a leading cause of childhood morbidity in sub-Saharan Africa. This study evaluated whether an Enhanced Seasonal Malaria Chemoprevention (SMC) protocol, combining standard antimalarials with high-dose iron and nutritional supplementation, produces superior haemoglobin trajectories compared to a Standard SMC protocol over six months during the rainy season. A simulated Randomised Controlled Trial (RCT) enrolled 300 children measured at months

0, 2, 4 and 6, adding to a total of 1,200 observations. Since repeated measurements within the same child were correlated (ICC=0.704), a Linear Mixed Model (LMM) was fitted. A random intercept and random slope model was selected as the final model over a random intercept only model via Likelihood Ratio Test ($\chi^2 = 10.24$, $df=2$, $p = 0.006$). The treatment $\times$ month interaction coefficient was $\hat{\beta}_3 = 0.205$ g/dL/month (95%CI: [0.177,0.232]; $p < 0.001$), corresponding to a ≈ 1.23 g/dL cumulative advantage for the Enhanced SMC arm by month 6. Diagnostic plots confirmed model assumptions were reasonably satisfied. The statistical evidence supports national scale-up of Enhanced SMC, subject to cost-effectiveness and equity analyses.

1. Introduction

Malaria is a primary cause of childhood anaemia in sub-Saharan

Africa through erythrocyte haemolysis, bone marrow suppression and iron sequestration, producing dangerous drops in haemoglobin (Hb), particularly during the rainy season. Standard Seasonal Malaria Chemoprevention (SMC) reduces clinical malaria incidence during the transmission peak but its haematological impact is limited, as it may not offset the nutritional deficits that accompany repeated infections. An Enhanced SMC protocol combining standard antimalarials with high-dose iron and nutritional supplementation may amplify haematological benefits by directly addressing the iron deficiency component of malaria-associated anaemia alongside controlling parasite burden. Since each child contributed four repeated measurements, observations within the same child are correlated, requiring a Linear Mixed Model (LMM) for valid statistical inference.

Primary research question: Does Enhanced SMC produce significantly better haemoglobin trajectories over six months than Standard SMC, after controlling for age and bed-net use?

Objectives: (i) Compare baseline characteristics to assess randomisation; (ii) visualise individual and population-level haemoglobin trajectories; (iii) build and select the best-fitting LMM; (iv) validate model assumptions through diagnostic plots; (v) translate statistical findings into a Ministry of Health recommendation.

2. Methodology

2.1 Study Design and Population

A simulated six-month longitudinal Randomised Controlled Trial (RCT) enrolled 300 children during the rainy season, randomised as follows:

- Standard SMC (Control, $n = 155$): standard antimalarial chemoprevention administered throughout the follow-up period.
- Enhanced SMC (Intervention, $n = 145$): standard antimalarials plus high-dose iron and nutritional supplementation.

Each child was assessed at four time points, months 0 (baseline), 2, 4 and 6, yielding 1,200 total observations. The primary outcome was haemoglobin concentration (g/dL). Covariates included age at enrolment (months) and insecticide-treated net use (ITN: Consistent or Inconsistent).

2.2 Statistical Methods

2.2.1 Exploratory Data Analysis

Baseline characteristics were compared between treatment arms using Student's t -test for continuous variables and Pearson's χ^2 test for categorical variables, summarised as Mean(SD) and N (%) respectively. Spaghetti plots with Locally Estimated Scatterplot Smoothing (LOESS)-smoothed population means and 95% confidence intervals, mean trajectory plots with error bars, and boxplots by time point were produced to visualise haemoglobin trajectories. Additional plots faceted by ITN use and age group were produced to explore the influence of these covariates.

All analyses were conducted in R version 4.5.2 using the packages **lme4**, **lmerTest**,

finalfit, **tidyverse**, **broom.mixed** and **kableExtra**.

2.2.2 Linear Mixed Model

The general LMM for individual i at occasion j is:

$$Y_{ij} = X_{ij}\beta + Z_{ij}b_i + \varepsilon_{ij} \quad (1)$$

where Y_{ij} is the vector of haemoglobin observations for child i ; X_{ij} is the design matrix of fixed effect predictors (month, treatment, age, ITN use); β is the vector of fixed effect coefficients; Z_{ij} is the design matrix for the random effects; $b_i \sim N(0, D)$ is the vector of random effects; and $\varepsilon_i \sim N(0, \sigma^2 I)$ is the vector of residual errors, with $b_i \perp \varepsilon_i$.

Four models were built sequentially:

Model 0 – Unconditional Means Model (no predictors):

$$Hb_{ij} = \beta_0 + b_{0i} + \varepsilon_{ij} \quad (2)$$

where β_0 is the overall mean haemoglobin across all children and time points, b_{0i} is the random intercept for child i , and ε_{ij} is the residual error. This model is used solely to partition variance and compute the Intraclass Correlation Coefficient (ICC):

$$ICC = \frac{\sigma_{b_0}^2}{\sigma_{b_0}^2 + \sigma_{\varepsilon}^2} \quad (3)$$

An $ICC > 0.05$ justifies the use of an LMM over standard linear regression.

Model 1 – Fixed effects, random intercept, no interaction:

$$Hb_{ij} = \beta_0 + \beta_1 \text{month}_{ij} + \beta_2 \text{Tx}_i + \beta_3 \text{age}_i + \beta_4 \text{ITN}_i + b_{0i} + \varepsilon_{ij} \quad (4)$$

where β_1 is the effect of time (month), β_2 is the effect of treatment arm, β_3 is the effect of age at baseline and β_4 is the effect of ITN use. This model introduces all main fixed effects with a random intercept only, with no interaction term, establishing the baseline relationship between predictors and haemoglobin before testing the interaction in Model 2.

Model 2 – Treatment $\times$ month interaction, random intercept:

$$Hb_{ij} = \beta_0 + \beta_1 \text{month}_{ij} + \beta_2 \text{Tx}_i + \beta_3 (\text{month} \times \text{Tx})_{ij} + \beta_4^{\text{age}} \text{age}_i + \beta_5 \text{ITN}_i + b_{0i} + \varepsilon_{ij} \quad (5)$$

The interaction β_3 is the primary coefficient of interest. A treatment main effect alone captures whether average haemoglobin differs between arms regardless of time, whereas the interaction specifically estimates the differential rate of haemoglobin change per month, that is, whether children in the Enhanced SMC arm gain haemoglobin faster than those in

the Standard SMC arm over time. This is the precise mechanism the intervention is designed to produce.

Model 3 – Random intercept + random slope for month:

$$b_i = \begin{pmatrix} b_{0i} \\ b_{1i} \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} \sigma_{b_0}^2 & \sigma_{b_{01}} \\ \sigma_{b_{01}} & \sigma_{b_1}^2 \end{pmatrix}\right) \tag{6}$$

where b_{0i} is the random intercept and b_{1i} is the random slope for month for child i ; $\sigma_{b_0}^2$ is the between-child variance in baseline haemoglobin; $\sigma_{b_1}^2$ is the between-child variance in individual rates of change over time; and $\sigma_{b_{01}}$ is their covariance. This model allows each child their own individual haemoglobin trajectory slope, capturing the reality that children differ not only in where they start but in how fast their haemoglobin changes over time. The full Model 3 specification is:

$$\begin{aligned} \text{Hb}_{ij} = & \beta_0 + \beta_1 \text{month}_{ij} + \beta_2 \text{Tx}_i \\ & + \beta_3 (\text{month} \times \text{Tx})_{ij} + \beta_4^{\text{age}}_i \\ & + \beta_5 \text{ITN}_i + b_{0i} + b_{1i} \text{month}_{ij} + \varepsilon_{ij} \end{aligned} \tag{7}$$

The Likelihood Ratio Test (LRT) was used to compare Model 2 against Model 3. lme4 automatically refits both models using maximum likelihood (ML) internally when anova() is called. The selected model was then refit using restricted maximum likelihood (REML) via update() for unbiased variance estimates. p values used the Satterthwaite degrees-of-freedom approximation (lmerTest).

3. Results

3.1 Baseline Characteristics

Table 1: Baseline characteristics by treatment arm. Mean(SD) for continuous; N (%) for categorical. p -values: t -test / χ^2 .

Characteristic	Total Participants, N (%)	Missing, N		Standard SMC	Enhanced SMC	p-value
Total Participants, N (%)				155 (51.7)	145 (48.3)	
Age at Baseline (months)	300 (100.0)	0	Mean (SD)	34.4 (12.4)	36.9 (13.3)	0.094
ITN Use	300 (100.0)	0	Inconsistent	56 (36.1)	73 (50.3)	0.018
			Consistent	99 (63.9)	72 (49.7)	
Haemoglobin (g/dL)	300 (100.0)	0	Mean (SD)	12.0 (0.9)	12.0 (1.0)	0.668

Note: Continuous variables: Mean (SD). Categorical: N (%). p -values: t -test (continuous), chi-squared (categorical). Non-significant p -values indicate effective randomisation. ITN = Insecticide-Treated Net.

Table 1 shows that age (p -value = 0.094) and baseline haemoglobin (p - value = 0.668) are well balanced between the two arms, confirming that randomisation was effective and that any haemoglobin differences observed during follow-up can be attributed to the intervention rather than pre-existing group differences. ITN use, however, differs significantly between arms (p - value = 0.018), with 63.9% of Standard SMC children reporting consistent use compared to 49.7% in the Enhanced SMC arm. This imbalance, while not the result of failed randomisation per se, justifies the inclusion of ITN use as a covariate in all subsequent models to control for its independent effect on haemoglobin.

3.2 Haemoglobin by Time Point

Table 2: Mean(SD) haemoglobin (g/dL) by treatment arm and time point.

Treatment	Month	N	Mean	SD	Min	Max
Standard SMC	0	155	11.98	0.94	9.2	14.3
Standard SMC	2	155	11.60	0.99	9.1	14.1
Standard SMC	4	155	11.29	1.03	8.4	13.8
Standard SMC	6	155	11.00	1.04	8.5	13.4
Enhanced SMC	0	145	12.03	0.96	9.9	14.6
Enhanced SMC	2	145	12.09	0.98	9.8	15.6
Enhanced SMC	4	145	12.21	0.99	10.0	15.1
Enhanced SMC	6	145	12.28	1.00	10.0	15.0

Table 2 shows both arms begin with nearly identical baseline haemoglobin (Standard: 11.98g/dL; Enhanced: 12.03g/dL). The Standard arm declines progressively to 11.00g/dL by month 6 (loss of $\approx$ 0.98g/dL), while the

Enhanced arm improves to 12.28g/dL, yielding a descriptive gap of $\approx$ 1.28g/dL before any formal modelling.

3.3 Exploratory Visualisations

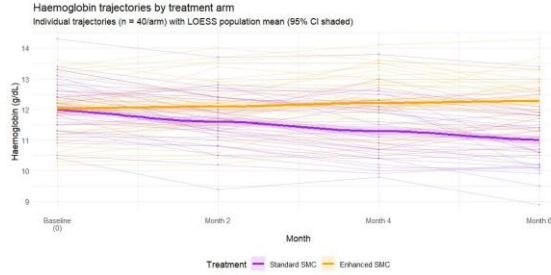


Figure 1: Individual haemoglobin trajectories ($n = 40$ per arm shown) overlaid with LOESS population mean and 95% CI from all children per arm.

Figure 1 displays individual trajectories for 40 children per arm overlaid with LOESS-smoothed population means and 95% CI.

Substantial between-child variability, with trajectories crossing and diverging widely, directly justifies the LMM over standard regression. Both mean lines start at identical positions at baseline, then diverge progressively: the Enhanced SMC mean rises while the Standard SMC mean declines, with clear separation by month 4 and month 6.

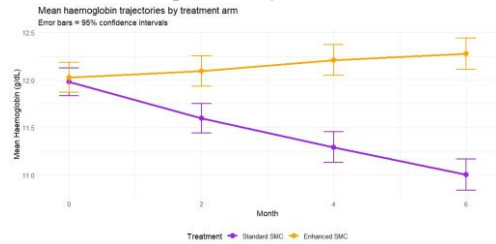


Figure 2: Mean haemoglobin trajectories by treatment arm with 95% CI error bars.

Figure 2 presents the mean haemoglobin trajectories for each arm with 95% confidence interval error bars at each time point. At baseline, the error bars for both arms overlap substantially, consistent with effective randomisation. By month 2 the confidence intervals begin to show some separation, and by month 4 and month 6 the intervals separate clearly and no longer overlap, providing strong descriptive evidence of a statistically meaningful divergence in haemoglobin trajectories between the two arms. The steady upward trajectory of the Enhanced SMC mean contrasted with the declining Standard SMC mean visually foreshadows the significant treatment $\times$ month interaction that formal modelling will confirm.

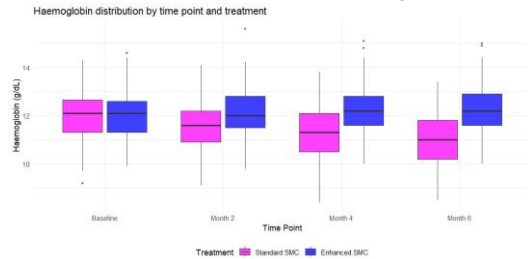


Figure 3: Haemoglobin distributions by time point and treatment arm.

Figure 3 presents side-by-side boxplots of haemoglobin at each of the four time points, grouped by treatment arm. At baseline (month 0), the two arms show nearly identical median values and interquartile ranges, confirming balance. From month 2 onwards, a clear separation in medians emerges, with the Enhanced SMC arm showing consistently higher median haemoglobin values and broadly stable spread. In contrast, the Standard SMC arm shows a progressive downward shift in both the median and the overall distribution at every subsequent time point. The absence of extreme outliers in either arm and the relatively stable spread within each arm over time suggests that the distributional assumptions required for linear modelling are plausible.

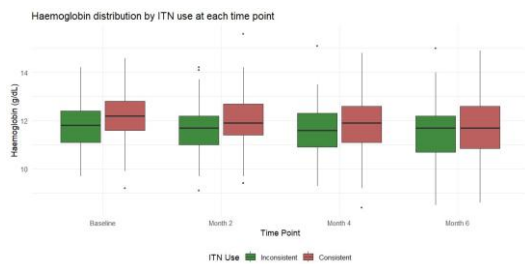


Figure 4: Haemoglobin distributions by ITN use at each time point.

Figure 4 shows haemoglobin stratified by ITN use at each time point, pooled across arms. Children with consistent ITN use maintain higher haemoglobin throughout all four time points including baseline, suggesting the protective effect is a stable pre-existing characteristic. This justifies adjusting for ITN use in all models, particularly given the baseline imbalance between arms.

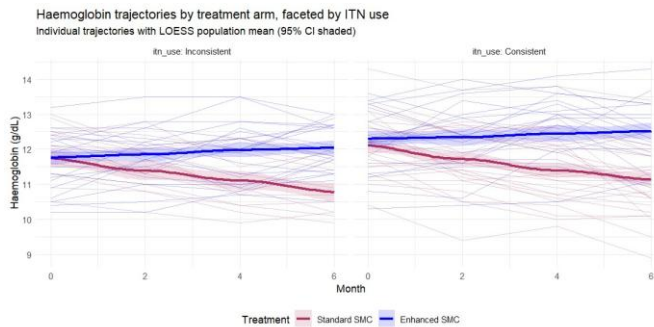


Figure 5: Haemoglobin trajectories faceted by ITN use (LOESS mean and 95% CI).

Figure 5 presents LOESS mean trajectories faceted by ITN use. The Enhanced SMC advantage is present in both the consistent and inconsistent subgroups, demonstrating that the treatment benefit is not confounded by bed-net use. Within each arm, consistent ITN users maintain a higher overall haemoglobin level, reinforcing the independent protective role of bed nets.

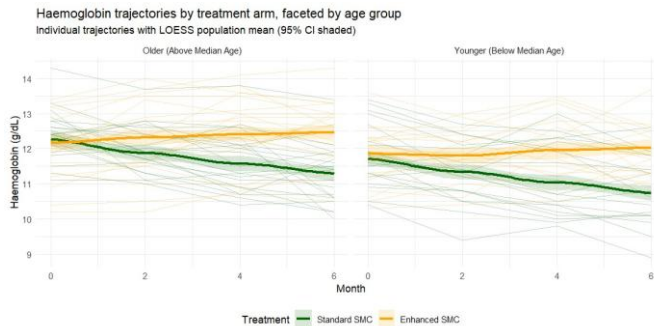


Figure 6: Haemoglobin trajectories faceted by age group (median split).

Figure 6 shows trajectories faceted by age group (median split).

The Enhanced SMC arm diverges upward relative to the declining Standard arm in both subgroups, indicating the treatment benefit is not restricted to a specific age band. This supports a universal targeting approach rather than restricting supplementation by age.

3.4 Model Building and Selection

3.4.1 Intraclass Correlation Coefficient

Table 3: Variance components and ICC from the unconditional means model (M0).

Variance Component	Estimate
Between-child variance (σ_b^2)	0.8234
Within-child variance (σ_w^2)	0.3456
Total variance	1.1690
ICC	0.7043

Note:
ICC = between-child variance / total variance. Values > 0.05 justify the use of a linear mixed model.

Table 3 presents the variance components from the unconditional means model. The ICC of 0.704 indicates that 70.4% of the total haemoglobin variance is attributable to stable between-child differences, that is, most of the variation in haemoglobin in this dataset comes from the fact that some children simply have higher haemoglobin than others throughout the

study, regardless of time or treatment. Only 29.6% of the total variance reflects withinchild change over time, which is the component the intervention is designed to target. An ICC this large justifies the use of an LMM over standard linear regression as applying standard regression would ignore this clustering, severely underestimate standard errors and produce test statistics and p-values that are misleadingly significant.

3.4.2 Model Comparison

Table 4: Model comparison: AIC, BIC, log-likelihood and LRT.

Model	Parameters	AIC	BIC	Log-Likelihood	LRT p-value
M2: Random Intercept Only	8	2470.47	2511.19	-1227.23	-
M3: Random Intercept + Random Slope	10	2464.23	2515.13	-1222.12	0.006

Note:
Lower AIC/BIC = better fit. LRT $p < 0.05$ indicates the random slope significantly improves model fit. Models estimated by maximum likelihood (ML) for comparison.

Table 4 presents the model comparison between Model 2 (random intercept only) and Model 3 (random intercept plus random slope for month). The Likelihood Ratio Test was significant ($\chi^2 = 10.24$, $df=2$, p -value = 0.006) and Model 3 also produced a lower AIC (2464.2 vs 2470.5), indicating that the addition of a random slope for month significantly improved model fit. Model 3 was therefore selected as the best and final model, confirming that children vary not only in their baseline haemoglobin levels but also in their individual rates of change over the six-month follow-up. The selected model was refit using REML for unbiased variance component estimation.

3.5 Model Diagnostics

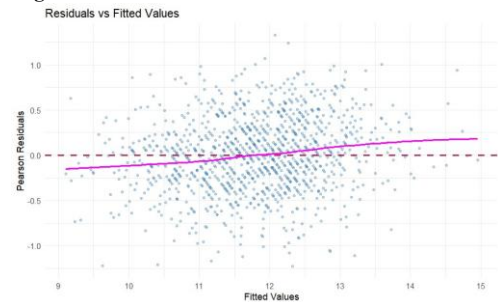


Figure 7: Residuals vs fitted values from the final model.

Figure 7 shows the Pearson residuals plotted against fitted values from the final model. The residuals are scattered approximately randomly around the zero reference line across the full range of fitted values, with no systematic fanning, curvature or pattern. The LOESS smoother tracks closely along the zero line throughout, confirming the absence of any systematic trend in the errors. These features together indicate that the homoscedasticity assumption is reasonably satisfied, that is, the model errors do not grow or shrink as a function of the predicted haemoglobin level.

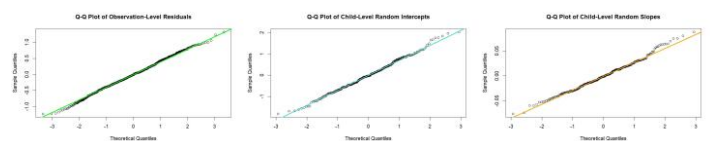


Figure 8: Q-Q plots: (left) observation-level Pearson residuals; (centre) child-level random intercepts; (right) child-level random slopes.

Figures 8 present Q-Q plots for the three sets of residuals from the final model. The observation-level Pearson residuals (left panel, $n = 1,200$) follow the reference line closely across the central quantiles, with only minor tail deviations typical of real-world data, confirming approximate normality of within-child errors. The childlevel random intercepts (centre panel, $n = 300$) align closely with the reference line across the full range of quantiles, confirming that individual deviations in baseline haemoglobin are approximately normally distributed. The child-level random slopes (right panel, $n = 300$) also follow the reference line closely, confirming approximate normality of individual rates of change. Taken together, all diagnostic plots support the conclusion that the assumptions of the final model are reasonably satisfied and that the statistical inference produced by the model is valid.

3.6 Confirmatory Analysis

3.6.1 Fixed Effects

Table 5: Fixed effects from the final LMM (REML). Reference category: Standard SMC, Inconsistent ITN use. p -values via Satterthwaite degrees-of-freedom approximation.

Term	Estimate	Std. Error	95% CI	p-value
Intercept	10.965	0.159	[10.653, 11.277]	< 0.001
Month	-0.162	0.030	[-0.181, -0.143]	< 0.001
Treatment: Enhanced SMC	0.078	0.101	[-0.121, 0.276]	0.443
Age at baseline (months)	0.021	0.004	[0.013, 0.028]	< 0.001
ITN use: Consistent	0.430	0.097	[0.239, 0.62]	< 0.001
Month x Treatment (Enhanced SMC)	0.205	0.014	[0.177, 0.232]	< 0.001

Note:
Reference: Standard SMC, Inconsistent ITN use. Highlighted row = primary coefficient of interest. p -values via Satterthwaite degrees-of-freedom approximation (lmerTest).

Table 5 presents the fixed effects from the final model. Each coefficient is interpreted below.

Intercept ($\hat{\beta}_0 = 10.965$, p -value < 0.001): This represents the estimated mean haemoglobin at baseline (month 0) for a child in the Standard SMC arm with inconsistent ITN use and mean age. It serves as the reference point from which all other effects are measured.

Month ($\hat{\beta}_1 = -0.162$, p -value < 0.001): In the Standard SMC arm, haemoglobin declined by 0.162g/dL per month, corresponding to a cumulative loss of approximately 0.97g/dL over the six months of follow-up. This statistically significant decline reflects ongoing malaria pressure and the absence of the additional nutritional supplementation provided in the Enhanced arm.

Treatment: Enhanced SMC ($\hat{\beta}_2 = 0.078$, p -value = 0.443,

NS): The non-significant treatment main effect indicates that there was no meaningful difference in haemoglobin between the two arms at baseline (month 0). This is consistent with effective randomisation and confirms that the two groups started from the same haemoglobin level before the intervention took effect.

Month×Treatment: Enhanced SMC ($\hat{\beta}_3 = 0.205$, 95% CI:

[0.177, 0.232], p -value < 0.001): This is the primary finding of the study. For each additional month of follow-up, children in the Enhanced SMC arm experienced a haemoglobin trajectory that was 0.205 g/dL per month higher than children in the Standard SMC arm. Over the full six months, this differential compounds to an estimated ≈ 1.23 g/dL cumulative advantage for the Enhanced arm. This is clinically meaningful: mild anaemia in young children is defined as haemoglobin below 11 g/dL, and a 1.23 g/dL improvement could represent the difference between a child remaining anaemic and recovering to a healthy haemoglobin level.

ITN: Consistent ($\hat{\beta}_4 = 0.430$, p -value < 0.001): Independent of treatment arm, children with consistent insecticide-treated net use had a haemoglobin level 0.430g/dL higher on average than those with inconsistent use. This large and highly significant independent effect supports the promotion of ITN use as a complementary strategy to SMC in national malaria prevention programmes.

Age at baseline ($\hat{\beta}_5 = 0.021$, p -value < 0.001): Each additional month of age at enrolment was associated with a 0.021g/dL higher haemoglobin level, reflecting the gradual maturation of haematopoietic capacity in young children as they develop. While the per-month effect is small, it is statistically significant and justifies age as a covariate adjustment in the model.

3.6.2 Random Effects

Table 6: Random effects variance components from the final model (REML).

Variance Component	Estimate
Between-child: Intercept variance ($\sigma_{b_0}^2$)	0.5832
Between-child: Slope variance ($\sigma_{b_1}^2$)	0.0031
Within-child: Residual variance (σ_e^2)	0.2291

Table 6 presents the random effects variance components. The large intercept variance ($\hat{\sigma}_{b_0}^2 = 0.583$) confirms substantial betweenchild differences in baseline haemoglobin, while the slope variance ($\hat{\sigma}_{b_1}^2 = 0.003$) confirms children also differ in their individual rates of change, which justifies Model 3 over Model 2. The residual variance ($\hat{\sigma}_e^2 = 0.229$) captures remaining within-child variability.

3.6.3 Model-Predicted Trajectories

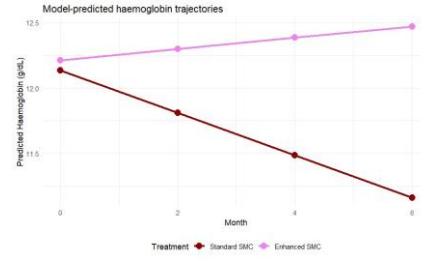


Figure 9: Model-predicted marginal haemoglobin trajectories from the final LMM, at mean age and consistent ITN use.

Figure 9 presents the model-predicted marginal haemoglobin trajectories at mean age and consistent ITN use. The Enhanced SMC arm rises steadily at $-0.162 + 0.205 = +0.043$ g/dL per month while the Standard SMC arm declines at -0.162 g/dL per month, producing a widening gap that reaches ≈ 1.2 g/dL by month 6 and directly visualises the cumulative benefit of the Enhanced SMC protocol.

4. Discussion

The treatment×month interaction ($\hat{\beta}_3 = 0.205$ g/dL/month, p -value < 0.001) provides robust statistical evidence that Enhanced SMC produces significantly better haemoglobin trajectories than Standard SMC over six months. This is biologically plausible: iron supplementation restores depleted iron stores while nutritional cofactors support erythropoiesis once antimalarials have reduced parasite burden. The non-significant treatment main effect (p -value = 0.443) confirms the advantage emerged over time through the intervention, not from pre-existing group differences.

The ICC of 0.704 validated the LMM framework and Model 3 was selected over Model 2, confirming that children vary in both baseline haemoglobin and individual rates of change. The significant protective effect of consistent ITN use ($\hat{\beta}_4 = 0.430$, p -value < 0.001) further supports bed-net promotion as a complementary strategy alongside SMC.

Limitations: The dataset is simulated, complete adherence was assumed, time-varying confounders were not modelled and longer follow-up would be needed to confirm whether the haemoglobin advantage persists beyond the rainy season.

5. Conclusion & Policy Recommendation

This study provides strong statistical evidence that Enhanced SMC produces significantly better haemoglobin trajectories over six months. The treatment × month interaction was highly significant (p -value < 0.001) with a clinically meaningful effect of 0.205 g/dL per month, amounting to a ≈ 1.23 g/dL cumulative advantage by month 6, and all model assumptions were satisfied.

Recommendation to the Ministry of Health: Based purely on the statistical evidence from this trial, yes, the Enhanced SMC protocol should be scaled up nationally. The treatment×month interaction was highly significant (p -value < 0.001), the effect size is clinically meaningful, model assumptions were satisfied and the benefit was consistent across both age groups and ITN use categories. We recommend national scale-up be accompanied by: (1) a formal cost-effectiveness analysis of the supplementation component; (2) a larger multi-site confirmatory trial to verify generalisability; and (3) sub-group analyses by baseline anaemia severity to optimise targeting.

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